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Meta-Learning Enhanced Explainability of Large Language Models for Low-Sample Transaction Monitoring in AML Compliance

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Abstract

In the atmosphere of the strict regulation of finance industry, AML compliance imposes higher demands on interpretability and identification attention of transaction monitoring systems under low-sample scenario, while the current rule-based systems have the limitation of data shortage and diversiform behavior patterns. Towards this end, in this work we introduce a novel and interpretable large language model framework Meta-XAML, enhanced by meta-learning, for more intelligent and transparent AML transaction monitoring in low-shot setting. On the one hand, the task adaptive meta-learner is capable of dynamically adapting the model parameters to the heterogeneity of customer behavior and trans-jurisdictional differences. Secondly, the attribution-guided interpretation module makes attention aligned with the causal mask mechanism of Shapley value, to achieve legally traceable interpretation for high-risk transactions. Furthermore, the model also formulates the objective of prototype constraint contrast learning for better semantic alignment and more robust expression under few shots. By jointly optimizing multiple objectives loss functions, the entire model is forced to be optimized to ensure: the classification accuracy, interpretation credibility, and cross-domain generalization ability. Experiments on the AML trading dataset demonstrate that Meta-XAML achieves 8.6% F1-score gain over the state-of-the-art competitors.

CCS Concepts

• **Security and privacy** → Human and societal aspects of security and privacy; Economics of security and privacy.

Keywords

Meta-Learning, Explainability, Large Language Models, Anti-Money Laundering, Few-Shot Learning

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1 Introduction

In the face of the ever-growing complex global interconnected financial system, the compliance of anti-money laundering (AML) has reached the top challenges for some financial institutions like banks, securities companies and payments platforms. Now more and more cross-border transactions and new payment methods including virtual currency and art transactions, money laundering behavior has a stronger concealment and the complexity of the structure [1]. Different from the traditional rule-based or expert rule-based monitoring system, it has been found that the existing system has such issues as lag of identification and high false positive ratio in dealing with strategies like the structured division transactions, rolling transfer and the mock legal route based on the international binary options trading. Facing the tide of RegTech innovation, it is imperative for financial institutions to implement intelligent algorithms with data-driven and adaptive learning as well as strong generalization to realize all-round progress of the real-time, sensitivity and compliance of transaction monitoring [2].

Recently, pre-trained large language models (LLMs) like T5, GPT, BERT, etc., have demonstrated great superiority in natural language understanding and generation. Leveraging v tat many semantics learning from large corpus, should extract insightful semantic connections among unstructured texts including transaction desc, customer profiling and transaction path that is why behavior inference complexities are enable to interpret complex behavior pattern and aid when suspecting on frauds at transaction level helps interpret complex behavior pattern at complex level, respectively [3]. In particular, in text heavy financial tasks, LLMs provide a strong modelling basis for 'suspicious but not explicit' transaction behaviour. Nevertheless, the "black box nature" of these architectures introduces some lack of interpretability that cannot be neglected. The basis of decision-making is not easy to trace, so the model existence the bottleneck of its promotion in the practical application of AML less legally feasible explanation in the compliance report, audit verification and even court statement.

Additionally, in practical application of AML transaction monitoring system, we are usually confronted with the problems of insufficient data labeling, and the serious imbalance between positive and negative samples. The vast majority of suspected transaction cases are not easy to establish a structured and high-quality large-sample data due to that they are highly confidential and hard to collect evidence. Such low-sample, or even zero-shot Gentysuch

world makes it unrealistic for classical supervised deep learning methods to work, trained the fast R-CNN model will work good on the training set, but easy to overfit and fail the prediction on the new world, with different natural conditions, for example geographic environment [4]. Accordingly, learning framework that can achieve high-performance prediction while possessing strong interpretability under the small number of labeled samples is the one of the major directions for the development of AML intelligent monitoring system.

Meta-learning, a research direction which has become increasingly vital in recent years, aims to simulate the learning mechanism of human beings' "deriving from one another's experience" and acquire the ability of fast adaptation and learning on the new task sequentially [5]. The intuition is to condense multiple similar learning tasks to share general optimization strategy or parameter initialization, thus enabling rapid fine-tuning on a target task, which significantly mitigates the generalization barrier due to data scarcity. In the sense of regulation, model decisions should not only be "effective", but also be "explainable". Financial regulators, the internal audit departments and legal review teams will not just want to know if a transaction is classified as suspicious, but "why it is deemed to be suspicious" and if there exists a quantifiable and auditable causal path.

2 Related work

Above all, Day [6] studies the strategies used for money laundering in the misuse of electronic payment instruments and cryptocurrencies to model the entities and relationships regarding money laundering, in order to support the effective use of AML in fintech. Cm et al. [7] tackles logical table-to-text generation challenge, in which models dynamically choose appropriate table attributes, while alleviating factual inconsistencies in attribute-to-value mapping (eg., from a table column with possible yes and no values to a text value like yes-No where-) as well as adaptation to newly added attributes with minimal retraining. Yang et al. [8] concentrates on finding laundering activities from the unlabeled data by creating explicit anomalous transaction data and considering complex transaction patterns.

Tang et al. [9] proposed an anti-money laundering method based on Hierarchical Risk Control Knowledge Graph (HRCKG). By extracting transaction features from simulated AML data, HRCKG was constructed to automatically identify and classify suspicious transaction behaviors. Yan et al. [10] investigated how large language models (LLMs) could improve financial regulation, especially in the context of anti-money laundering. Traditional AML technology is difficult to adapt to complex and large financial transaction data since the procedure of using rule-based mechanism is mostly of manual work.

Rani et al. [11] presented a spatiotemporal network driven approach to analyze bank transaction behavior for detecting possible money laundering cases in order to increase the efficiency and accuracy of detecting money laundering activities. By building a spatiotemporal network model, the temporal and spatial features of transaction behaviors are characterized to facilitate the discovery of suspicious activities more efficiently, and improve the efficiency and efficacy of the AML (anti-money laundering) system. Li [12] talked about using Graph Neural Network (GNN) for several tasks, and particularly for AML tasks. The study suggests that two distinct neural networks are trained for anti-money laundering to enhance the performance and interpretability of the model to make it more suitable to the demand of financial compliance.

3 Methodologies

3.1 Adaptive meta learner

Meta-XAML uses a pre-trained language model (such as T5 or FLAN-T5) as the underlying encoder to map the context information of each transaction to a high-dimensional semantic embedding space. In order to adapt to the heterogeneity of customer behavior and different judicial supervision scenarios, the model introduces a task-level meta-learner TAML to extract initialization parameters with generalization ability from multiple tasks, so that the model can quickly complete adjustment and discrimination with only a few observations.

We represent each transaction as an input sequence x_i with a label $y_i \in \{0, 1\}$ to indicate whether it is suspicious or not. With *LLM* f_θ, it can be mapped as a high-dimensional semantic vector $h_i \in \mathbb{R}^d$, as shown in Equation 1:

$$h_i = f_\theta\left(x_i\right), \tag{1}$$

where θ is the learnable parameter of the language model. This embedding representation will serve as the basis for subsequent interpretation generation and classification. To model the meta-learning process, we consider the task set $\mathcal{T} = \{\mathcal{T}_1, \mathcal{T}_2, \ldots, \mathcal{T}_K\}$, where each task contains a support set $\mathcal{S}_k$ and a query set $\mathcal{Q}_k$. TAML uses gradient descent to generate task-specific parameters in each task, as Equation 2:

$$\theta_k^{'} = \theta - \alpha \nabla_\theta \mathcal{L}_{\mathcal{S}_k}\left(f_\theta\right), \tag{2}$$

where $\mathcal{L}_{\mathcal{S}_k}$ is the loss function (e.g., cross-entropy) on the support set, ∇_θ is the gradient computation of the original parameters, and α is the learning rate. The formula shows that in the support set of each task, the model is updated in an internal loop according to the current parameter θ, and the step size is α, and the parameter $\theta_k^{'}$ adapted to the current task is obtained. Subsequently, TAML minimizes the loss on the query set for all tasks and optimizes the model's initialization parameter θ, as shown in Equation 3:

$$\theta^* = arg \min_\theta \sum_{\mathcal{T}_k \sim \mathcal{T}} \mathcal{L}_{\mathcal{Q}_k}\left(f_{\theta_k^{'}}\right), \tag{3}$$

The formula shows that the model optimizes the initial parameter θ^* by quickly adapting to multiple tasks and evaluating the performance on the query set, so as to improve its cross-task generalization and migration capabilities.

The AGEM module first calculates the original prediction score using the attention matrix as shown in Algorithm 1, then applies Shapley approximation and causal mask to each token to measure its contribution, and finally fuses the two to generate a fine-grained attribution score to visually reflect the impact of each token on the model output.

3.2 Explainability enhancement mechanisms

In order to meet the requirements of financial compliance, the model must not only make accurate judgments, but also provide

Algorithm 1 Pseudo-Code for Attribution-Guided Explanation Module.

Input: Transaction text tokens, Fine-tuned LLM M with attention A, Shapley approximation function $\phi(\cdot)$.
Output: Attribution scores.
1. Compute attention matrix A = M.compute_attention($\{w_j\}$)
2. score_orig ← M.predict($\{w_j\}$)
3. for each token w_j do
   Shapley-like attribution
   Causal masking verification
   mask_input = mask(w_j)
   score_mask = M.predict(mask_input)
   Δ_j = |score_orig − score_mask|
   $\alpha_j = combine(\alpha_{sj}, \Delta_j)$
 end for
return set of α

clear, causal, and legally explanatory visual outputs. To this end, Meta-XAML introduces the Attribution-Guided Explanation Module (AGEM), which combines Shapley value-based attention attribution and causal masking mechanism to generate structured and traceable explanations. When predicting whether a transaction is suspicious, it is especially critical for the model to focus on which fragments of the input text are present. We represent the attention matrix in the LLM as Equation 4:

$$A_i = Attn\left(x_i; \theta\right). \tag{4}$$

AGEM calculates the attribution importance score for each word using a Shapley-like value method, as shown in Equation 5:

$$\phi_j\left(x_i\right) = \mathbb{E}_{S \subseteq \mathcal{V} \setminus \{j\}}\left[Score\left(S \cup \{j\}\right) - Score\left(S\right)\right], \tag{5}$$

where $\mathcal{V}$ is the set of input words, and $Score(S)$ is the suspicious score output by the model when only a subset of words S is retained.

The formula estimates the causal effect of the j-th word by calculating the marginal contribution of the jth word to the prediction score of the model under all possible subsets of words.

To further verify the causal significance of the attribution score, AGEM performed a masking experiment on each word. Specifically, the model output is re-predicted after the j-th word is masked, and the prediction difference is calculated, as shown in Equation 6:

$$CausalEffect_j = \hat{y}_{full} - \hat{y}_{masked(j)}, \tag{6}$$

This value reflects the actual intervention role of the word j in the final prediction. If the difference is large, it means that the word has a significant causal impact on the model prediction, and can be used as the compliance basis for the determination of suspicious transactions.

To enhance the semantic alignment ability in low-shot scenarios, we introduce a prototype contrast learning mechanism to constrain the samples to be closer to the center of their respective categories. Let each category $c \in \{0, 1\}$ correspond to the archetype representation $p_c \in \mathbb{R}^d$, then the contrast loss for each sample x_i is Equation 7:

$$\mathcal{L}_{Proto} = -log\frac{\exp\left(sim\frac{\left(h_i, p_{y_i}\right)}{\tau}\right)}{\sum_{c'}\exp(sim\frac{\left(h_i, p_{c'}\right)}{\tau}}, \tag{7}$$

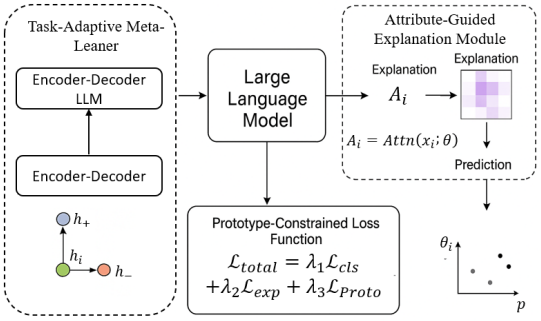

Figure 1: Meta-XAML Framework Architecture Diagram.

where $sim(a, b)$ is the cosine similarity function, and τ is the temperature coefficient, which controls the smoothness of the distribution. This formula is based on the $softmax$ normalized form, which brings the samples closer to the corresponding class prototype and away from other classes, improving the expression consistency of fewer samples. The final training goal of Meta-XAML consists of three parts: classification accuracy, interpretation consistency, and semantic alignment, as shown in Equation 8:

$$\mathcal{L}_{total} = \lambda_1\mathcal{L}_{cls} + \lambda_2\mathcal{L}_{exp} + \lambda_3\mathcal{L}_{Proto}, \tag{8}$$

where $\mathcal{L}_{cls}$ is the binary cross-entropy loss, $\mathcal{L}_{exp}$ measures the consistency of the interpretation result with the manual annotated attribution, $\mathcal{L}_{Proto}$ is the comparison loss of the aforementioned prototype. λ_1, λ_2 and λ_3 is the hyperparameter weight. Through multi-objective optimization, the model can achieve good performance under limited data, while ensuring the traceability and semantic rationality of the output interpretation.

Figure 1 illustrates the overall flow of Meta-XAML in a low-sample AML transaction monitoring task. Based on the pre-trained Encoder–Decoder large language model, the task adaptive meta-learner module generates task-specific parameters from a few-shot support set and then inputs them into the core Large Language Model. The attribution-guided interpretation module generates a traceable explanation of each transaction by extracting the attention matrix and combining causal masking.

4 Experiments

4.1 Experimental setup

The experiment was tested with the Synthetic Anti-Money Laundering Dataset (SAML-D) [13], which includes over 1 million bank transactions, consisting of structured fields such as transaction amount, account ID, transaction type, and whether it was flagged. SAML-D is grounded on realistic financial behavior simulation, achieving strong reproductions, and low positive rate of 1.5%, consistent with data scarcity and data imbalance in AML scenarios. The following are four typical methods used in this study to compare with the proposed Meta-XAML model:

- eXtreme Gradient Boosting (XGBoost) is a classical tree model based on gradient boosting, it is a popular choice

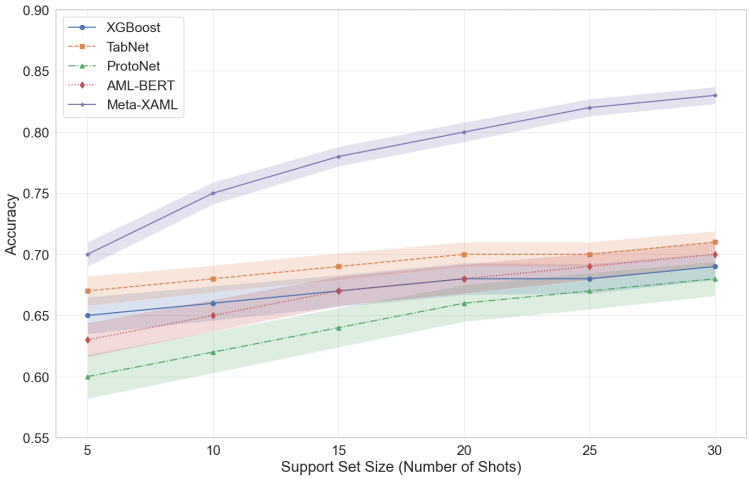

Figure 2: Accuracy Comparison Across Methods in Low-Sample AML Monitoring.

for both financial fraud detection and anti-money laundering because of its remarkable classification performance and robustness.

- Tabular Deep Learning Network (TabNet) is a deep neural network model for structured data modeling that combines interpretability and end-to-end modeling. The model can capture high-order interaction features of transactions by the attention mechanism, and achieves good results of financial risk control and fraud detection.
- Prototypical Networks for Few-Shot Learning (ProtoNet) is a paradigm of such a primitive few-shot meta-learning technique that performs category matching and categorization by creating "prototype representation" of each class. The algorithm is rooted from the trans-action representation, and the decision is conducted in the nearest neighbor prototype distance way, but without the need of countless training samples.
- Pretrained Language Model (AML-BERT) is an AML model constructed by directly fine-tuning a pre-trained language model like BERT. The transaction description is retrofit to nature language input in this approach, and the classification plane is learned with a finetune way.

4.2 Experimental analysis

As shown in Figure 2, the accuracy of the five methods all follow an upward direction as the support set size increases, which is also evidence that more few-shot data would improve the learning of the model. In their work, the performance of XGBoost and TabNet does not grow much faster compared to other models: in XGBoost, the performance grows from about 0.65 to about 0.71 when increasing the size of the support set from 5 to 30. It was the lowest at small samples for ProtoNet (≈ 0.60), however, it showed a significant improvement with the increasing number of samples, approximately 0.68 at 30, and text features also allowed AML-BERT to obtain an accuracy among TabNet and Meta-XAML, approximately 0.70 at 30 samples. On the other hand, Meta-XAML exhibited the highest initial value (≈ 0.70) and also the greatest increment, from 0.70 at 5 to 0.83 at 30 samples and always with a lead values.

As we can observe from Table 1, the F1 values of all methods are increased gradually when using more and more training data (from 10% to 90%), which shows that the more data the model used, the better its performance. In the four among them, XGBoost, TabNet, ProtoNet and AML-BERT, the F1 values were relatively low (approximately 0.58–0.67) with low data proportions (10%–30%), and when the data proportions were moderate (50%–70%), TabNet and AML-BERT began to significantly surpass XGBoost and ProtoNet. On the contrary, Meta-XAML has always taken the lead in the whole scale, with an F1 value changing from 0.68 to 0.85 as data ratio scales from 10% to 90%, and the margin is the largest (0.17), suggesting that Meta-XAML has strong learning ability under low-sample conditions, as well as further cement the lead with the increase of data amount, which verifies the reliability and stability of the meta-learning and prototype comparison mechanism towards small-shot AML tasks.

The confidence of the interpretations of all methods decreases with the perturbation ratio between 10% and 50% and it obvious from Figure 3 that more input features are blanked out, smaller the effect of the model interpretation on the prediction. XGBoost and ProtoNet were consistently the worst in all levels of perturbation, with an original fidelity of ~0.15 and ~0.12 respectively, then decreased to ~0.11 and ~0.08 at 50%, TabNet, AML-BERT were in the middle (TabNet got down to ~0.18 from ~0.14, AML-BERT from ~0.16 to ~0.12), and Meta-XAML maintained the highest fidelity (from ~0.20 to ~0.16) in all levels, and experienced the smallest drop (the explanation reflected the prediction behavior of the model's best, had the strongest causal consistency and robustness).

5 Conclusion

The Meta-XAML framework developed in this paper naturally integrates TAML, AGEM and prototype constraint comparison to enable superior-effective classification and high-quality interpretability in

Table 1: F1 Score Comparison.

Training Data Proportion (%)	XGBoost	TabNet	ProtoNet	AML-BERT
10	0.62	0.64	0.58	0.61
20	0.64	0.66	0.6	0.63
30	0.65	0.67	0.62	0.65
40	0.66	0.68	0.63	0.66
50	0.67	0.69	0.64	0.67
60	0.68	0.7	0.65	0.68
70	0.68	0.71	0.66	0.69
80	0.69	0.71	0.67	0.7
90	0.7	0.72	0.68	0.71

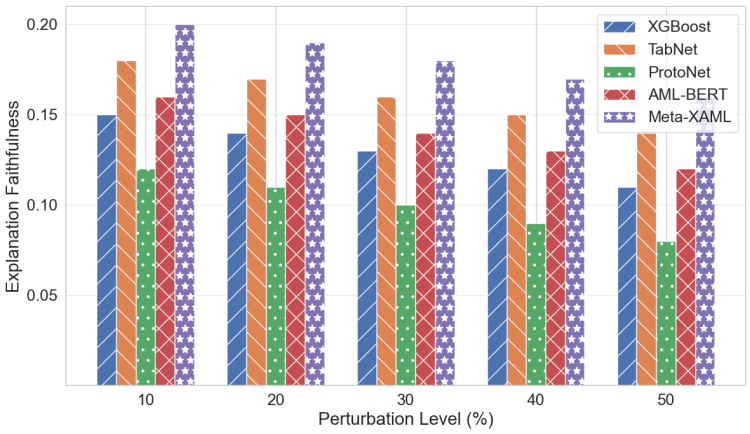

Figure 3: Explanation Faithfulness Comparison Across Methods.

low-sample anti-money laundering transaction monitoring. Specifically, we propose a task-adaptive transfer learning (TAML) module for fast adapting pre-trained gL2RLMs to new tasks, AGEM for a judicial traceable attention explanation path for suspicious transactions by Shapley-like value attention attribution and causal masking, prototype comparison loss for improved semantic fidelity and robust representations in few-shot settings. The results of experiments indicate that both on accuracy, F1value and interpretability of the models. As for future improvements, Meta-XAML can expand in the following two aspects: On the one hand, we aim to apply Meta-XAML to more real and diversified cross-border financial datasets to investigate the generalization ability, and on the other hand, we could introduce more fine-grained causal reasoning or visualization techniques into the interpretation module for interpretation at ease and for legal illustration.

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